PROJECT PROGRESS

Hot Hand, Real or Believed?

A behavioural analysis of setting decisions in professional volleyball

PROGRESS UPDATE

Studies on Human Behaviour · Project exam

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Project recap

RQ1

Real hot hand

Does an attacker's previous success predict her next-attack success?

Performance → performance

RQ2

Believed hot hand

Is the same attacker more likely to be chosen again after a successful attack?

Performance → decision

RQ3

Competitive pressure

Does competitive pressure change attack performance or attacking choices?

Pressure → behaviour

DATASET COLLECTED

Data collection completed

63	36	27
match files, total	Brescia matches	Talmassons matches

2 shared head-to-head matches

Matches cover different phases of the 2025-26 season

Regular season	Coppa Italia	Promotion play-offs

Parsing

1

Team identification

The '*'/'a' marker identifies home and away teams, but Brescia and Talmassons switch sides across matches.

A fixed home/away rule would assign the focal team inconsistently across the season.

Teams are identified directly by team name, using an is_own_team flag

2

Library compatibility

The PyPI release of pydatavolley is not fully compatible with the NumPy and pandas versions available in Colab.

NumPy 2.0: error mixing text and missing values.

pandas 3.0: chained .fillna(..., inplace=True) no longer supported.

Compatibility fixes are applied before parsing, without changing the library’s parsing logic

Checking the reconstructed data

1

Score reconstruction

Reconstructed running score was compared with the official set-by-set results stored in each file header.

Scores matched, once accounting for the different recording of the final point of some sets

2

Match structure

Number of reconstructed sets per match.

All 63 matches contained 3, 4, or 5 sets, consistent with the best-of-five format

3

Player identity

Player identification across own-team attacks.

8,312 own-team attacks linked to a valid attacker

23 distinct attackers identified across both squads

RQ1 — is there an early signal of a real hot hand?

Descriptive comparison (raw rates)

BRESCIA

+3.4 pp

Attack success is 3.4 percentage points higher when the same attacker's previous attempt was successful.

TALMASSONS

+1.7 pp

The same pattern appears, but the difference is smaller than for Brescia.

At this stage, these are descriptive differences only

PRELIMINARY RESULTS

RQ2 — does previous success affect the setter’s next choice?

Descriptive comparison (raw rates)

BRESCIA

≈ 0 pp

The probability of choosing the same attacker again is almost unchanged after success.

TALMASSONS

−6.0 pp

The same attacker is re-selected less often after a successful attack.

These are raw rates only, these differences still needs to be tested with the mixed-effects model

Could the apparent hot-hand effect be due to player differences?

Attackers have different baseline probabilities of success because of differences in ability and playing role.

Could this create an apparent hot-hand pattern even without any real within-player effect?

NEXT STEP

Mixed-effects logistic regression

Account for repeated observations and differences between players and matches.

Random intercepts: attacker – match

Main effect of interest: previous attack success

RQ3: Competitive pressure

RQ3 requires two additional variables not directly available in the parsed data: attack tempo and competitive pressure.

1 Attack tempo

Decode the attack-combination codes into meaningful tempo categories.

TEMPO CATEGORIES

Quick – Super – Medium – High – Other

2 Competitive pressure

Use the running score to identify attacks performed under higher competitive pressure.

CRITICAL PHASE

Sets 1–4: either team has reached 20 points

Set 5: the entire set is considered critical

NEXT STEPS

What remains to be done

1

Test RQ1 and RQ2

Fit the mixed-effects models and check whether the preliminary patterns remain after accounting for player and match differences

→

2

Complete RQ3 variables

Decode attack tempo and classify critical vs non-critical phases

→

3

Test RQ3

Compare attack performance and tempo under different pressure conditions

→

4

Final interpretation

Bring the three research questions together and discuss the results in behavioural terms

Where the project stands

✓

DATA COLLECTION

complete

63 match files collected

✓

DATA PREPARATION

complete

Parsing issues addressed and main validation checks completed

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ANALYSIS

in progress

Mixed-effects models and RQ3 still to be completed

Preliminary descriptive results show different patterns across the two research questions and between the two teams. The next step is to test whether these patterns remain after accounting for player and match heterogeneity.

Next step

Testing the preliminary patterns with the mixed-effects models